

AP CSP | Lesson 27: Impacts & Digital Divide

Homework | Multiple-choice and written practice

Name: _____ Class: _____ Date: _____

Time: 40 minutes | Total: 30 teacher marks (not an AP score) | Show working.

Original AP-style MCQs and classroom written practice; these are not official AP questions.

Q5: select exactly TWO. All case numbers and costs are synthetic. Explain mechanisms and show calculations.

1. A person has a working laptop. Which statement is justified? [1 mark]

- | | |
|-----------------------------------|---|
| A. Internet access is guaranteed. | B. Device access and Internet access are separate requirements. |
| C. Bandwidth must be unlimited. | D. All websites must be accessible. |

Answer: _____

2. In a fictional map project, more visitors may increase shop sales and nearby noise. This illustrates that: [1 mark]

- | | |
|--|---|
| A. Every effect is harmful. | B. Only creators are stakeholders. |
| C. One effect can be beneficial and harmful. | D. Unintended effects cannot be beneficial. |

Answer: _____

3. Creators intend a fictional map to support shade planning. Residents later use it to plan art walks. The new use is: [1 mark]

- | | |
|---|--|
| A. Necessarily harmful. | B. Proof the original purpose failed. |
| C. Impossible to anticipate in principle. | D. An unintended use that may be beneficial. |

Answer: _____

4. Which is an example of a digital divide issue? [1 mark]

- | | |
|---|---|
| A. Two connected users prefer different colors. | B. Two programs use different variable names. |
| C. One area lacks affordable Internet access available elsewhere. | D. Two files have different extensions. |

Answer: _____

5. Select exactly TWO justified statements. [1 mark]

- | | |
|--|--|
| A. Skill training can address a practical participation barrier. | B. A free device removes all possible access barriers. |
| C. Only governments can affect the digital divide. | D. Access differences can occur locally and globally. |

Answer: _____

6. A model says 85 people can access a fictional system. Which claim is justified? [1 mark]

- | | |
|--|---|
| A. Exactly 85 people will submit real reports. | B. The model enables access for 85 under its assumptions. |
| C. Every resident benefits equally. | D. The model proves real policy costs. |

Answer: _____

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Written responses | Questions 7-9

7. [4 marks]

A fictional shade map makes neglected places more visible. Explain one possible benefit and one possible harm, naming the affected stakeholder and mechanism for each.

8. [4 marks]

A fictional map intended for planning is reused for art walks. Classify this new use, explain a possible benefit, state one responsible design action and one prediction limit.

9. [4 marks]

Synthetic groups: 40 have no barrier, 20 need connection only, 15 device only, and 10 both. All other people stay blocked. Give model access with no action, connection only, device only and both; explain the combined case.

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Written responses | Questions 10-12

10. [4 marks]

Synthetic baseline: 40. Exclusive groups need connection only 20, device only 15, skills only 10, accessibility only 5, both connection/device 10. Costs: connection/device 2 each; skills/accessibility 1 each. Budget 4. Compare connection+device with connection+skills+accessibility: costs, totals, remaining groups, and one evaluation limit.

11. [4 marks]

In a fictional town, a free hotspot is open only during office hours. Some residents work then; others need an accessible interface. Explain the barrier the hotspot addresses, a time limit, another matched action and a success measure.

12. [4 marks]

Evaluate: "A fictional model enables all 100 residents, so all will contribute and benefit equally." Identify two unsupported claims, propose one field measure and explain how that measure could guide revision.
